

Peter Racioppo

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Machine Learning Engineer and researcher specializing in Transformer architectures, sequence modeling, and dynamical systems. Solo author of an ICML 2026 Spotlight paper and first author on multiple machine learning publications, with experience training language models, developing novel attention mechanisms, and building training/eval infrastructure.

EXPERIENCE

Machine Learning Research Intern | Doheny Eye Institute 2024 – 2026

- Developed a knowledge distillation framework for retinal image segmentation across multiple ocular diseases.
- Developed a novel Sliding Window Attention mechanism that imposes temporal stationarity priors for robust learning in low-data regimes; validated on longitudinal retinal disease forecasting.
- First author on two peer-reviewed medical AI publications. ([MDPI Bioengineering 2025](#), [MDPI Life 2026](#)).

Research Assistant | Center for Vision, Cognition, Learning, and Autonomy (UCLA) 2023 – 2025

- Developed novel Transformer architectures grounded in dynamical systems, Bayesian filtering, and state estimation, culminating in a solo-authored ICML 2026 Spotlight paper.
- Built PyTorch training infrastructure and conducted large-scale language-model experiments to evaluate long-context extrapolation and architectural design choices.

RESEARCH HIGHLIGHTS

Robust Filter Attention: Self-Attention as Precision-Weighted State Estimation. P. Racioppo. [[Paper](#) | [GitHub](#)]
International Conference on Machine Learning (ICML, 2026) – Spotlight

- Solo-authored ICML 2026 Spotlight paper, deriving a new class of attention mechanisms from state-estimation principles and demonstrating improved language-model perplexity relative to RoPE across long-context benchmarks.
- Built and trained Transformer language models end-to-end, conducting extensive ablations and maintaining stable zero-shot extrapolation at up to 8× training context length.
- Open-sourced implementation with reproducible experiments.

The Transformer as a Polar State Estimator. P. Racioppo. [[Preprint](#)]

- Showed that attention, residual connections, and normalization can be derived as a first-order approximation to a geometric state estimator on the hypersphere, giving a probabilistic interpretation of the Transformer architecture.
- Proposed the Polar Transformer, a new architecture retaining higher-order geometric corrections and introducing magnitude-aware attention precision.

SELECTED PROJECTS

Hallucination Detection for LLMs. [[GitHub](#)]

- Developed a correctness-estimation system that predicts LLM answer reliability from internal Transformer activations, attention patterns, and residual stream dynamics.
- Built an adaptive retrieval framework that selectively triggers and evaluates RAG responses based on predicted correctness, improving the accuracy–efficiency tradeoff.

AC-SINDy: Compositional Sparse Identification of Nonlinear Dynamics. [[Preprint](#) | [GitHub](#)]

- Developed a scalable method for learning symbolic models of dynamical systems with improved robustness to noise.
- Implemented models and training pipeline and evaluated on simulated dynamical systems.

SKILLS

Languages: Python, R, MATLAB, SQL.

Core Areas: Sequence modeling, NLP, computer vision, dynamical systems, deep learning, statistical modeling.

ML & Deep Learning: PyTorch, scikit-learn, Transformers, CNNs, PEFT/QLoRA, model training/finetuning, evaluation.

Systems & Tools: Git, CUDA/GPU training, Hugging Face, cloud training, Pandas, NumPy, Jupyter.

EDUCATION

MS in Statistics, UCLA – 2025

MS in Mechanical Engineering (Controls), UCLA – 2023

BS in Physics, UCLA